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Title of Invention

System and Method for Indirect Estimation of Solar Irradiance and Heat Flux via Differential Thermodynamic Sensing

Application Information

APPLICATION TYPE	Utility - Provisional Application under 35 USC 111(b)	PATENT #	-
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PATENT CENTER #	70896421	AUTHORIZED BY	-
CUSTOMER #	–	FILING DATE	12/29/2025

Note Regarding Documentation:

This Electronic Receipt serves as official verification of the Priority Date (December 29, 2025) and Application Number (63/937,986). Due to standard international mailing timelines, the physical hard-copy is currently in transit to Sri Lanka, and a scanned copy will be uploaded immediately upon delivery. If the Admissions Committee requires direct government authentication, a Certified Copy as Filed can be commissioned directly from the United States Patent and Trademark Office (USPTO). For current evaluation purposes, the Application Number and Filing Date listed above constitute binding legal proof of the invention's filing status.

SYSTEM AND METHOD FOR INDIRECT ESTIMATION OF SOLAR IRRADIANCE AND HEAT FLUX VIA DIFFERENTIAL THERMODYNAMIC SENSING

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] Not Applicable.

TECHNICAL FIELD

[0002] The present invention relates generally to the field of environmental monitoring and embedded systems. More specifically, the present invention relates to systems and methods for estimating solar irradiance and heat flux using differential thermodynamic sensing and temporal derivative signal processing on standard capacitive environmental sensors.

BACKGROUND OF THE INVENTION

[0003] Current environmental monitoring infrastructure relies heavily on common capacitive sensors (e.g., Bosch BME280) that measure static variables: temperature, pressure, and humidity. These sensors typically answer static questions such as "What is the temperature?" or "How humid is it?" However, they remain unresponsive to the dynamic energy exchanges that dictate these conditions. Specifically, they fail to measure Global Horizontal Irradiance (GHI), the amount of solar energy reaching the ground per unit area, and Heat Flux, the rate of thermal energy transfer through the environment. These metrics are not merely abstract; they are essential for practical applications ranging from climate modeling to calculating evapotranspiration in agriculture.

[0004] The Hardware Limitation Traditionally, measuring solar radiation requires a thermopile pyranometer governed by ISO 9060 standards. These instruments operate on a logic comprising a black absorbing surface and a white reference surface, measuring the temperature difference via the Seebeck effect to generate a voltage proportional to incident radiation. While accurate for

research-grade applications, high-quality pyranometers are cost-prohibitive, often costing hundreds or thousands of dollars. They require careful mounting, cleaning, and calibration, and are fragile. Consequently, they are unsuitable for widespread deployment in high-density IoT networks or consumer electronics.

[0005] The Soft-Sensing Limitation: Recent attempts to estimate radiation using "soft sensing" algorithms, particularly machine learning (ML), have failed to provide robust solutions for embedded systems. Neural networks trained in specific climates often fail to generalize to new environments. Furthermore, the matrix multiplications required for inference consume excessive power on 8-bit microcontrollers. Critically, standard ML models often map "instant inputs" to "instant outputs," ignoring the temporal thermal inertia of the sensor itself. Since radiation changes on timescales of seconds (e.g., cloud movement) while sensor temperature changes on timescales of minutes, instantaneous ML models introduce significant errors.

SUMMARY OF THE INVENTION

[0006] To address the aforementioned limitations, the present invention provides a "Differential Temporal Derivative Soft-Sensing" (DTDSS) framework. This physics-based approach enables standard environmental sensors to function as radiometric instruments.

[0007] In a preferred embodiment, the system utilizes a paired-sensor configuration comprising a Reference Node and a Flux Node. By applying an Inertial Noise Reduction (INR) algorithm to the differential signal between these nodes, the system mathematically reconstructs solar energy intensity (Global Horizontal Irradiance) and heat flux. This method allows for precision environmental monitoring without the cost or fragility associated with specialized scientific hardware. By calculating complex energy flows using common sensors, the system opens a new frontier of applications, from precision engineering to automated energy management.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a cross-sectional schematic view of the Differential Sensing Head, illustrating the separation of the Reference Node and Flux Node.

[0009] FIG. 2 is a block diagram illustrating the signal processing pathways, including the Reference Path and Reactive Path.

[0010] FIG. 3 is a diagram illustrating the Proximity Correlation Hypothesis regarding wind speed in uniform environments.

[0011] FIG. 4 is a graph showing the comparative analysis of Solar Irradiance between the present invention (DTDSS) and a reference meteorological pyranometer.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

1. System Architecture

[0012] The present invention utilizes a differential thermodynamic approach, abbreviated as DTDSS (Differential Temporal Derivative Soft-Sensing). A single sensor cannot simultaneously achieve thermal equality with the air (for temperature/humidity) and thermal disequilibrium with solar radiation (for measurement). Therefore, the sensing architecture partitions the system into two separate thermodynamic nodes.

[0013] As illustrated in FIG. 1, the system utilizes two identical sensor elements placed in close physical proximity but in contrasting enclosures. The Reference Node comprises a sensor element (10) (e.g., a digital hygrometer) mounted on a substrate (12) and disposed within a ventilated enclosure (11). This enclosure (11) allows for free airflow (convection) while shielding the sensor (10) from direct radiation, ensuring it remains in thermal equilibrium with the ambient air mass.

[0014] Conversely, the Flux Node comprises a secondary sensor element (15) housed within a sealed chamber (17). This chamber acts as an energy trap. It is capped by a transparent cover (16) (e.g., glass or polycarbonate) which permits solar radiation to strike a black absorber plate (14) positioned directly above the sensor (15). A thermal housing separator (13) thermally decouples the two chambers to prevent conductive crosstalk. The separation is critical to prevent the "Dry Kiln" effect; by segregating the hygrometric measurement to the ventilated Reference

Node (10, 11), the system maintains psychrometric accuracy even while the Flux Node (14, 15, 17) reaches high stagnation temperatures.

[0015] Signal Processing Pathways: As shown in FIG. 2, the system employs a pipeline architecture. The raw thermodynamic data is ingested from the Reference Input (21) and the Flux Input (23). These signals enter the Processor (22) (which may be a microcontroller or edge-gateway).

The processing logic proceeds in sequential stages:

- Density & Enthalpy Calculation (Stage 22): The system first computes the air density and specific enthalpy using the reference pressure and humidity data.
- Differential Calculation (Stage 24): The system computes the raw delta

  ($T_{flux} - T_{ref}$).

- INR Filtering (Stage 25): The Inertial Noise Reduction filter is applied to the differential signal to recover high-frequency transients.
- Radiation Output (Stage 26): The final computed Solar Irradiance (W/m^2) is output for downstream applications.

[0016] Reference Path Logic (The Cloud Proxy Model): In a preferred embodiment, the Reference Path utilizes a 'Cloud Proxy Model' to establish a dynamic baseline for solar probability. The system recognizes that high surface relative humidity (RH) correlates with atmospheric opacity and cloud formation. The processor calculates a theoretical clear-sky radiation (G_{clear}) based on geolocation and time, and subsequently applies a cloud cover fraction (n) derived from the measured humidity:

$n = (RH/100)^p$ where p is a tuning parameter (typically 1.5 to 2.0).

Utilizing a modified Kasten-Czeplak relationship, the system calculates a maximum expected radiation baseline (G_{ref}):

$$G_{ref} = G_{clear} \times (1 - A \times n^B)$$

Where A and B are empirical coefficients (e.g., $A = 0.75, B = 3.4$). This baseline (G_{ref}) serves as a bounding constraint for the Reactive Path.

[0017] Bidirectional Parameter Adaptation: Unlike conventional soft-sensors, the processing pathways operate with bidirectional feedback. In the preferred embodiment, if the Reactive Path detects a sustained high-flux event that exceeds the Reference Path's predicted baseline (G_{ref}), the processor dynamically updates the cloud proxy tuning parameter

(p) in the Reference model. This allows the system to "learn" the local relationship between humidity and atmospheric opacity over time, effectively reducing the error residuals of the Kasten-Czeplak approximation.

2. Thermodynamics of the Medium (Moist Air)

[0018] Understanding the medium in which the sensor operates is essential. The sensor is fully immersed in the lower atmosphere, and the properties of that atmosphere directly govern heat transfer. A main innovation of the DTDSS framework is the rejection of static assumptions in favor of dynamic, first-principles derivation of air properties from the measured state variables: Pressure (P), Temperature (T), and Relative Humidity (RH).

[0019] Rejection of Standard Atmosphere Assumptions: Conventional environmental models often rely on "Standard Atmosphere" constants (e.g., sea level density

$\rho \approx 1.225$kg/m³). The present invention explicitly rejects these static assumptions. As demonstrated in field testing, a static density model deployed at 4,000 meters elevation results in a convective cooling overestimation of approximately 50%. By deriving density dynamically from real-time pressure (Stage 22 in FIG. 2), the DTDSS framework eliminates this systemic error without requiring external GPS altitude data.

[0020] Vapor Pressure Derivation: To determine density dynamically, the system first calculates the partial pressure of water vapor (e) using the measured Relative Humidity (RH) and Temperature (T in °C). The system applies the Magnus-Tetens approximation:

$$e = 6.112 \cdot \exp\left(\frac{17.67 \cdot T}{T + 243.5}\right) \cdot \frac{RH}{100}$$

With e resolved, the system computes the moist air density (ρ_{moist}) using the specific gas constant for dry air (R_d):

$$\rho_{moist} = \frac{P}{R_d(T + 273.15)}\left(1 - \frac{e}{P}(1 - \epsilon)\right)$$

Where $\epsilon \approx 0.622$.

By calculating ρ_{moist} dynamically at every timestep, the soft sensor automatically recalibrates its heat transfer coefficients for the exact altitude and humidity conditions, eliminating the need for manual altitude configuration.

[0021] Thermodynamic Distinction (Advective vs. Radiative): The system distinguishes between Radiative Heating (solar input) and Advective Heating (warm air mass movement) by monitoring the Specific Enthalpy (h) of the moist air. The processor calculates h using the mixing ratio x: $h = 1.006T + x(2501 + 1.86T)$

A rise in sensor temperature accompanied by a simultaneous rise in ambient specific enthalpy indicates Advective Heating (warm, moist air moving in). A rise in sensor temperature without a corresponding change in the ambient air's mixing ratio indicates Radiative Heating.

3. Mathematical Derivation of Heat Flux via Thermal Inertia

[0022] The approach treats the Flux sensor package as a Lumped Capacitance System (valid when Biot number < 0.1). Applying the conservation of energy to the Flux Node (S_{flux}):

$$(\alpha G A_s + P_{elec}) - h_c A_s (T_{flux} - T_{ref}) = m C_p \frac{dT_{flux}}{dt}$$

Where α is solar absorptivity, G is solar radiation,

P_{elec} is electrical self-heating, h_c is the convective heat transfer coefficient, and mC_p is the heat capacity.

[0023] Convective Physics: The convective heat transfer coefficient h_c is not a static constant but is derived from the Nusselt number (Nu), which relates convective to conductive heat transfer. The system models the sensor package as a flat plate in laminar flow, where $Nu \propto Re^{0.5} Pr^{0.33}$.

Since the Reynolds number (Re) is linearly dependent on air density (ρ), the system dynamically scales h_c as a function of the calculated ρ_{moist}. This ensures that the heat flux estimation remains accurate even as the convective cooling efficiency drops at high altitudes.

[0024] Solving for Solar Radiation: Rearranging the energy balance equation to solve for Solar Radiation (G):

$$G = \frac{h_c}{\alpha}\left[(T_{flux} - T_{ref}) + \tau \frac{dT_{flux}}{dt} - T_{rise}\right]$$

Here, we define two characteristic parameters:

Thermal Time Constant (τ): $\tau = \frac{mC_p}{h_c A_s}$.

This is the time it takes for the sensor to respond to a step change (typically 10s to 60s).

Self-Heating Offset (T_{rise}):

$$T_{rise} = \frac{P_{elec}}{h_c A_s}.$$

4. Inertial Noise Reduction (INR) Signal Processing

[0025] Taking the derivative ($\frac{dT}{dt}$) of a digital signal is prone to quantization noise. Standard filters (Moving Average) introduce delays that decouple the temperature from its derivative. The

present invention employs an Inertial Noise Reduction (INR) filter (Stage 25 in FIG. 2) operating in two stages:

[0026] Phase-Lead Compensation: The mathematical derivation of the temporal projection effectively functions as a digital phase-lead compensator. By injecting the derivative term ($\tau \cdot dT/dt$) into the calculation, the processor effectively predicts the future equilibrium temperature of the sensor before the physical mass has successfully heated. This transforms the naturally lagging physical system (a first-order thermal low-pass filter) into a zero-latency measuring device, allowing for the real-time recovery of instantaneous flux magnitude despite the thermal inertia of the physical enclosure.

[0027] Stage 1: Adaptive Smoothing. The system employs an Exponential Moving Average (EMA) where the smoothing factor α is not constant, but adapts dynamically to the volatility of the signal. The processor computes the deviation $\Delta[n]$ between the current raw sample and the previous filtered estimate. The smoothing factor is updated as:

$$\alpha[n] = \text{clamp}(k \cdot |\Delta[n]|, \alpha_{min}, \alpha_{max})$$

Where k is a sensitivity gain. During steady states (low deviation), α decreases to maximize noise rejection. During transient heating events (high deviation), α increases to minimize lag.

[0028] Stage 2: Inertial Projection. Following smoothing, the system computes a 'Projected Temperature' (T_{proj}) to compensate for the physical thermal lag of the sensor package:

$$T_{proj} = T_{filt} + \tau \cdot \frac{dT_{filt}}{dt}$$

This projection mathematically reconstructs the theoretical temperature the sensor would possess if it had zero thermal mass.

[0029] Advanced Derivative Computation: While the preferred embodiment utilizes a Central Difference method for computational efficiency on 8-bit cores, alternative embodiments utilizing 32-bit processors (e.g., ARM Cortex-M) may employ a Savitzky-Golay filter (polynomial least-squares smoothing). This method fits a low-degree polynomial to a sliding window of data

points (e.g., 5-10 samples) and calculates the analytical derivative at the center point, offering superior noise rejection for high-frequency sampling applications.

5. Environmental Boundary Conditions (The Wind)

[0030] Wind speed controls the convective heat transfer coefficient (h_c). The differential architecture employs Differential Cancellation: "Common mode" disturbances, like a sudden gust of wind, affect both T_{flux}andT_{ref} similarly. When the difference is calculated, wind noise subtracts out.

[0031] Proximity Correlation Hypothesis: As illustrated in FIG. 3, the system relies on a correlation between the local measurement site and the surrounding environment. The sensor unit (43), situated in a local environment (47), is subject to a local wind flow vector (48). Simultaneously, a distant reference area (44), containing landmarks (50), is subject to a regional wind flow (49).

The Proximity Correlation Hypothesis (46) posits that in uniform environments (where separation distance (45) is reasonably open), the magnitude of the local wind (48) approximates the regional wind (49) ($V_{local} \approx V_{regional}$). This allows the convective losses measured at the sensor (43) to represent the broader environmental state.

[0032] Hybrid Data Fusion: While the Proximity Correlation Hypothesis holds for open outdoor environments, indoor or complex urban microclimates may violate this assumption. In an alternative embodiment, the processor is configured to ingest external meteorological data (e.g., regional wind speed, cloud cover) via a network interface (e.g., LoRaWAN, Wi-Fi). The system fuses this external data with the local sensor readings to dynamically adjust the convective heat transfer coefficient (h_c), providing a residual correction that compensates for failures in the proximity assumption.

6. Hardware Implementation & Calibration

[0033] Dual Compartment Enclosure: The preferred embodiment utilizes a dual compartment design (FIG. 1). Compartment (11) (Reference) is a fully enclosed, white/reflective cylinder

allowing free air circulation but blocking direct line-of-sight to the sky. Compartment (17) (Flux) is topped with a transmissive material (16) and backed by a black surface (14).

[0034] Material Selection for Thermal Response: In a preferred embodiment, the absorptive enclosure of the Flux Node is constructed from a material having high thermal conductivity and high emissivity, such as anodized aluminum (e.g., 6061 alloy) with a black coating having an emissivity (ϵ) greater than 0.9. However, the invention is not limited to metal; alternative embodiments may utilize UV-stabilized dark polymers or carbon-loaded ABS, provided the thermal mass is characterized. The transmissive top cover is preferably composed of Borosilicate glass, Polycarbonate, or PTFE, selected based on the desired spectral cut-off frequencies for the target radiation (e.g., Shortwave vs. Longwave IR).

[0035] Memory Footprint Breakdown: The memory footprint is minimized by strictly managing state retention to fit within Class-0 device constraints. For the Reactive Path, the system requires only approximately 18 bytes of RAM per sensor node, distributed as follows:

Current & Previous Temperature: 4 bytes

Previous Derivative: 2 bytes

Smoothing Factor (α): 2 bytes

Circular Buffer (5 samples): 10 bytes

The complete differential thermodynamic state machine, including the Reference Path's enthalpy counters, occupies less than 60 bytes of total RAM.

[0036] System Calibration: The system is configured to perform a self-calibration routine to determine the specific thermal time constant (τ) of the Flux Node. The processor monitors for a 'Step-Down' event (e.g., sunset) and fits the decay profile to the natural cooling law:

$$T(t) = T_{final} + (T_{start} - T_{final})e^{-t/\tau}$$

[0037] Self-Heating Characterization: The parameter T_{rise} is non-zero due to the internal power dissipation of the sensor element. To determine this offset, the system utilizes a stored Lookup Table (LUT) correlating sampling rates (e.g., 1Hz vs 10Hz) to temperature offsets.

7. Experimental Validation

[0038] To evaluate performance, the DTDSS prototype was co-located with a calibrated standard pyranometer station. Data was logged synchronously over a complete diurnal cycle.

[0039] Comparative Results: FIG. 4 illustrates the raw thermodynamic response of the system (Solid Line) overlaid against reference meteorological data (Dashed Line). As predicted by the Lumped Capacitance model, the raw sensor signal exhibits a positive baseline offset during nocturnal periods (approximately 30-40 units), which corresponds to the Self-Heating Offset (T_{rise}) derived in Section 3. Despite the high-frequency convective noise visible in the raw signal, the system successfully captures the temporal phase of solar events (sunrise, solar noon, sunset). This raw data confirms the sensitivity of the differential architecture and validates the necessity of the Inertial Noise Reduction (INR) filter to extract the final calibrated irradiance from this noisy thermodynamic baseline.

8. Alternative Embodiments

[0040] Scaling and Accumulative Sensing: The differential principle described herein is not limited to a binary sensor pair. In an alternative embodiment, an array of N sensors is deployed across a structure (e.g., a building façade). The processor calculates thermal potentials between multiple reference points rather than a single differential.

[0041] Physics-Guided Machine Learning: In a hybrid embodiment, the physics-based DTDSS framework serves as the primary estimator, while a lightweight machine learning model (e.g., TinyML) is trained to predict the model residuals (errors).

[0042] Networked Fleet Calibration: In a deployed array of DTDSS nodes, the system employs spatial statistical analysis to identify sensor drift. If a single node reports a flux value significantly deviating from the spatial mean of its neighbors, the system flags the unit for "Soft-Recalibration."

[0043] Multi-Modal Sensor Fusion: The differential thermodynamic signals may be augmented by auxiliary low-cost sensors to resolve ambiguity. In one embodiment, a MEMS accelerometer is integrated to detect physical oscillation (wind sway). In another embodiment, a narrowband photodiode provides a binary "Day/Night" gate signal.

[0044] Fault-Tolerant Offline Logging: To ensure data integrity, the system comprises a "Heartbeat Monitor." Upon detection of network failure, the system automatically transitions to "Offline Mode," serializing data into a ring buffer.

[0045] Distributed Processing Architecture: The term 'processor' as used herein refers to any computational element, including local microcontrollers, edge gateways, or remote cloud servers. In a distributed embodiment, the sensor nodes transmit raw thermodynamic frames (Temperature, Humidity, Pressure) to a remote server, where the Differential Processing Engine executes the INR filtering and Flux inversion logic. This architecture allows for the deployment of ultra-low-power "dumb" sensors while maintaining the high-fidelity physics modeling centrally.

CLAIMS

What is claimed is:

1. A system for estimating solar irradiance and heat flux, comprising:
   a reference sensor node (10) disposed in a ventilated enclosure (11) configured to measure ambient thermodynamic parameters;
   a flux sensor node (15) disposed in a sealed, absorptive enclosure (17) configured to act as a thermal energy trap; and
   a processor (22) configured to:
   (a) calculate a differential temperature signal between the flux node and the reference node;
   (b) apply a dynamic noise reduction filter (25) to said signal, wherein the filter modulates a smoothing coefficient in response to a detected volatility metric of the signal; and
   (c) compute incident radiation by combining the differential temperature and a temporal projection derived from the rate of change of said differential temperature.
2. The system of claim 1, wherein the detected volatility metric comprises the rate of change of the acceleration (jerk) of the temperature signal.
3. The system of claim 1, wherein the processor is further configured to calculate the specific enthalpy of the ambient air using the humidity data from the reference node, and to inhibit solar radiation reporting when a temperature rise correlates with a rise in specific enthalpy (indicating advective heating).
4. The system of claim 1, wherein the processor calculates a local air density value derived from a pressure reading, and dynamically scales a convective heat transfer coefficient used in the flux calculation, thereby rendering the system altitude-independent.
5. The system of claim 1, wherein the processor is configured to execute an auto-calibration routine that detects a natural temperature decay event (sunset) and calculates a thermal time constant (τ) based on the decay duration.
6. The system of claim 1, wherein the sealed enclosure of the flux node (17) comprises a thermal insulator (13) positioned beneath the flux sensor node to minimize conductive heat loss, and wherein the enclosure is constructed of a material selected from the group consisting of anodized aluminum, UV-stabilized polymer, and carbon-loaded plastic.

7. The system of claim 1, wherein the processor is further configured to receive external meteorological data via a network interface and fuse said external data with local sensor readings to adjust a convective heat transfer coefficient.
8. The system of claim 1, wherein the INR filter utilizes a Savitzky-Golay polynomial least-squares smoothing algorithm to compute the derivative of the temperature signal.
9. The system of claim 1, wherein the processor is further configured to establish a bidirectional feedback loop wherein a high-flux event detected by the flux node dynamically updates a cloud proxy parameter in the reference node model.
10. The system of claim 1, further comprising an array of N sensors, wherein the processor computes statistical outliers among the array to perform networked self-calibration.

ABSTRACT OF THE DISCLOSURE

A system and method for estimating solar irradiance and heat flux using differential thermodynamic sensing on standard environmental sensors. The system employs a reference sensor in a ventilated enclosure and a flux sensor in a sealed, absorptive enclosure. A processor calculates the temporal derivative of the temperature difference between the nodes to reconstruct radiative flux. An Inertial Noise Reduction (INR) filter compensates for sensor thermal inertia and quantization noise, allowing for rapid detection of transient solar events. The system further calculates local air density from pressure readings to automatically adapt to varying altitudes. This approach enables radiometric monitoring using low-cost, hardware-independent components suitable for embedded IoT applications.

FIG 2

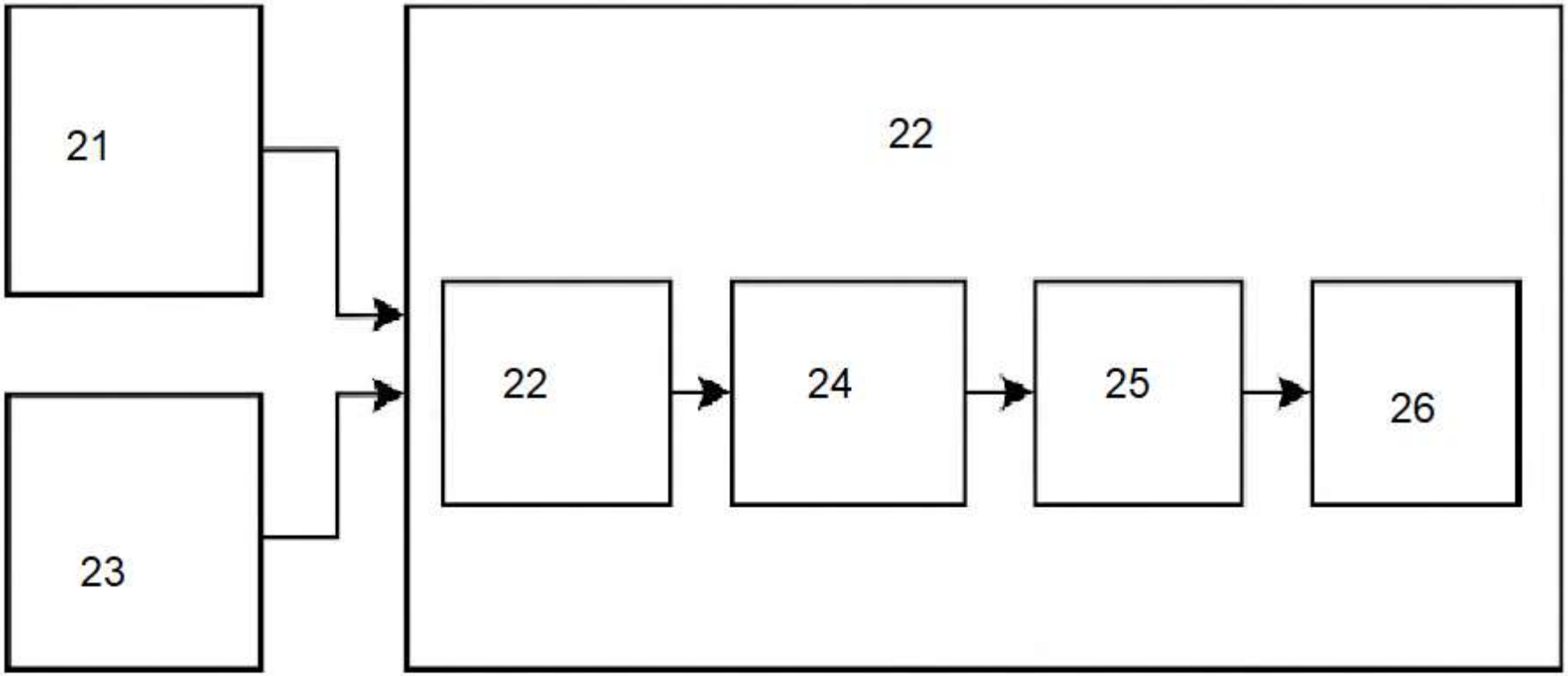


FIG 1

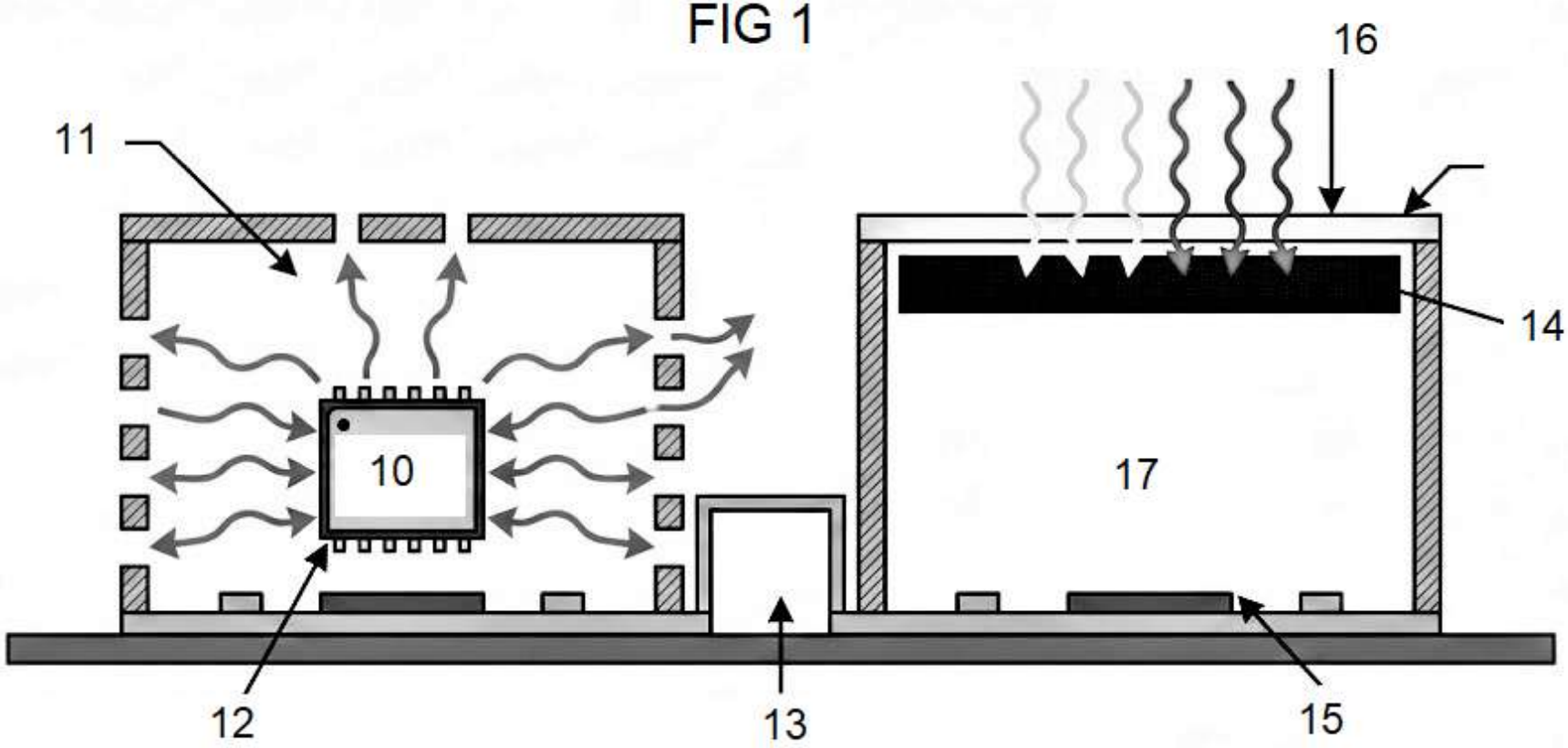


FIG 3

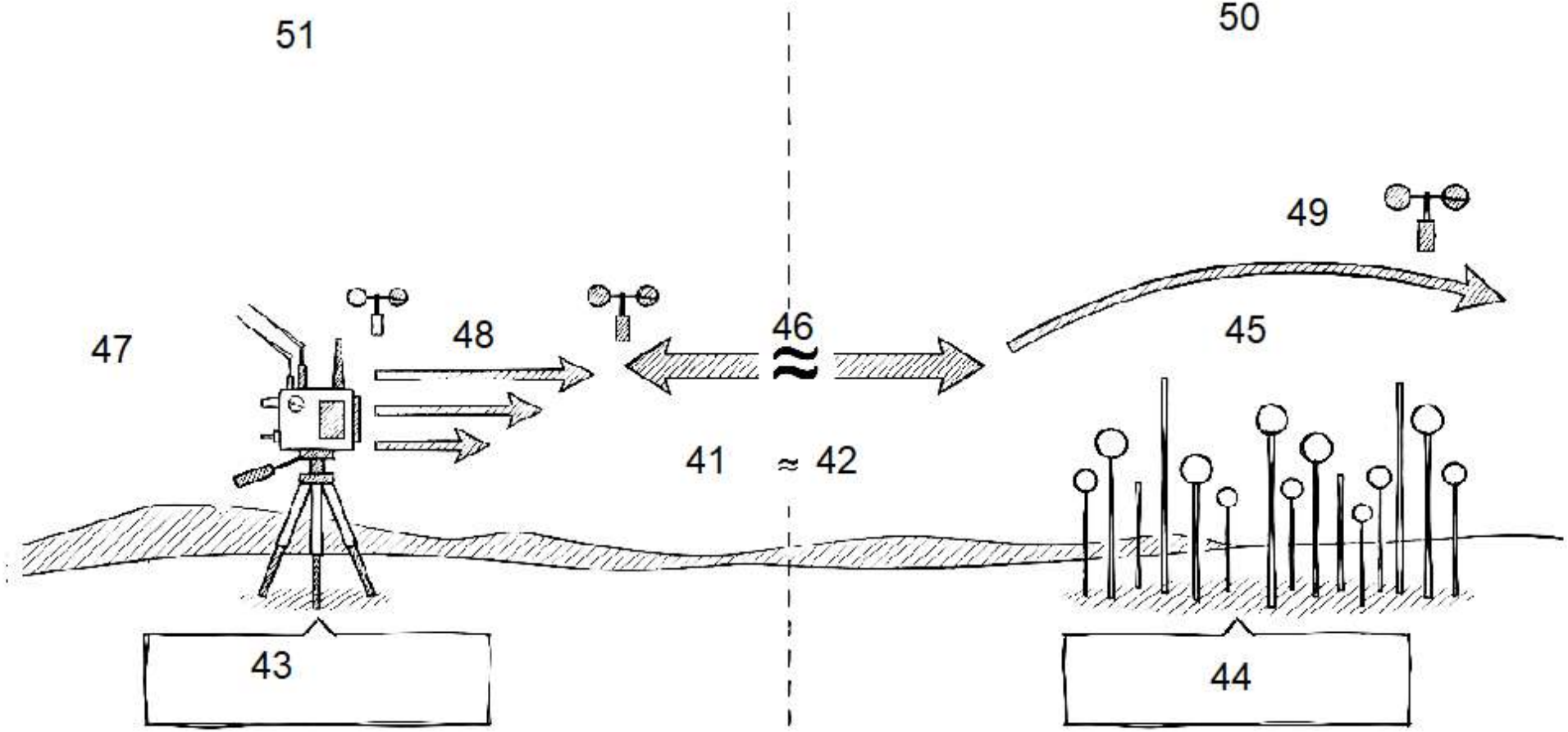
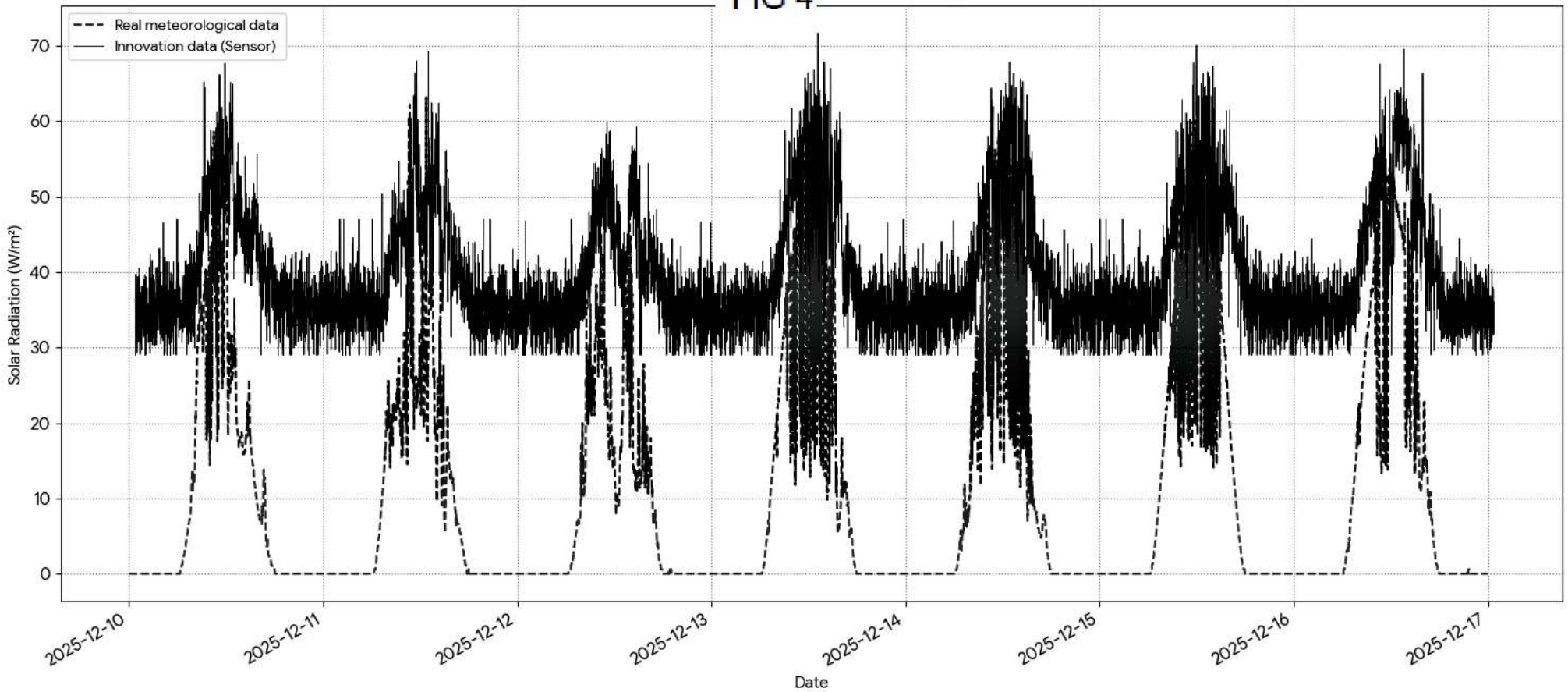


FIG 4


Real meteorological data
Innovation data (Sensor)
Solar Radiation (W/m²)
70
60
50
40
30
20
10
0
2025-12-10
2025-12-11
2025-12-12
2025-12-13
2025-12-14
2025-12-15
2025-12-16
2025-12-17
Date